Assessment – 5

Course Name : Analytical Bioinformatics Lab (Slot : L49+L50)

Course Code : BBIT417P

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Q1. A 34-year-old man comes to the clinic after two days of watery diarrhoea, abdominal

cramps, and a low-grade fever that began the morning after a backyard barbecue. He recalls

the potato salad had been sitting out in the sun most of the afternoon before he ate two helpings.

Several other guests reported similar symptoms over the following days. A stool sample was

collected and sent for 16S rRNA gene PCR and sequencing. The result:

> Patient_stool_isolate | partial 16S rRNA gene sequence | Sanger read

CTGATGGAGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAACGTCGCAA

GACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGATT

Based on the type of sequence you have been given, choose the appropriate BLAST program

and the most suitable database for identifying the pathogen, then perform the similarity search.

Part of the assessment is selecting the correct tool and database yourself.